

human

Show me the buildings within 5km of the building where OSID = '502c37bc-408b-4156-ac74-8d07946cffbe in Exeter

host_agent

Tool call: generate_metadata_for_all_artifacts with args: {} Tool response: {}

Tool call: send_message with args: {'target': 'named_area', 'task_description': 'Find the polygon for Exeter'}

named_area_agent

Tool call: call_os_ngd with args: {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_polygon'} Tool response: Search results have been found. Artifact generated is : exeter_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the polygon for Exeter., count of records fetched : 98.

Tool call: send_message with args: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_polygon'."}

data_analysis_agent

Tool call: generate_metadata_for_artifacts with args: {'artifact_names': ['exeter_polygon']} Tool response: Metadata generated for artifact 'exeter_polygon'.

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_polygon(data: list):
    gdf = data[0]
    # Filter for polygons specifically representing Exeter
    # Use name1_text for 'Exeter', and also check for descriptiongroup and description for city/town/settlement
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter', na=False))) &
                  (
                      (gdf['descriptiongroup'].str.lower().isin(['settlement'])) |
                      (gdf['description'].str.lower().isin(['city', 'town']))
                  )]
    return [
```

```
f"Filtered to {len(filtered)} polygon(s) for Exeter.",  
'filtered_exeter_polygon',  
'GeoDataFrame containing only the polygon(s) specifically representing Exeter, filtered by name and settlement/city/town type.',  
filtered  
]
```

Tool response: Filtered to 1 polygon(s) for Exeter.

Output: Filtered to 1 polygon specifically representing Exeter. - Artifact name: filtered_exeter_polygon - Description: GeoDataFrame containing only the polygon(s) specifically representing Exeter, filtered by name and settlement/city/town type.

named_area_agent

Output: The specific polygon for Exeter has been filtered and saved as: - **Artifact Name:** filtered_exeter_polygon - **Description:** GeoDataFrame containing only the polygon(s) specifically representing Exeter, filtered by name and settlement/city/town type.

Tool call: send_message with args: {'target': 'buildings', 'task_description': "Find the building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' within Exeter"}

buildings_agent

Tool call: call_os_ngd with args: {'filters': None, 'bbox': 'Exeter', 'polygon_or_point': None, 'street_address': None, 'filename': 'exeter_all_buildings'} Tool response: There was an error while calling NGD. Most probable cause is wrong artifact name for bbox parameter. Try a few times with the correct artifact names.

Output: I am unable to proceed because the required artifact name for the Exeter area (bbox parameter) is missing or incorrect. Please provide the correct artifact name for the Exeter area, or let me know if you would like me to list available artifacts to help identify the correct one.

Tool call: send_message with args: {'target': 'buildings', 'task_description': "Find the building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' within filtered_exeter_polygon"}

Tool call: call_os_ngd with args: {'filters': None, 'bbox': 'filtered_exeter_polygon', 'polygon_or_point': None, 'street_address': None, 'filename': 'exeter_buildings_all'} Tool response: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : ['buildings_exeter_buildings_all', 'buildingline_exeter_buildings_all', 'buildingpart_exeter_buildings_all'], Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Find the building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' within

filtered_exeter_polygon using filters None.", ...], counts of records fetched are : [69169, 69342, 85480].

Tool call: send_message with args: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'buildings_exeter_buildings_all' to find the building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe'. Save the result as 'building_with_osid_502c37bc.csv'."}

data_analysis_agent

Tool call: generate_metadata_for_artifacts with args: {'artifact_names': ['buildings_exeter_buildings_all']} Tool response: Metadata generated for artifact 'buildings_exeter_buildings_all'.

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_building_by_osid(data: list):
    # data[0] is the buildings_exeter_buildings_all GeoDataFrame
    gdf = data[0]
    filtered = gdf[gdf['osid'] == '502c37bc-408b-4156-ac74-8d07946cffbe']
    # Save to CSV (geometry will be WKT)
    filtered.to_csv('building_with_osid_502c37bc.csv', index=False)
    summary = f"Filtered to {len(filtered)} building(s) with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe'."
    artifact_name = 'building_with_osid_502c37bc.csv'
    artifact_description = "CSV file containing the building entry from 'buildings_exeter_buildings_all' with the specified OSID. Geometry is in WKT format."
    return [summary, artifact_name, artifact_description, filtered]
```

Tool response: The search return no results and no artifacts are generated

Output: No building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' was found in the provided dataset. No results were generated.

buildings_agent

Output: No building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' was found within the filtered_exeter_polygon area. There are no results matching this OSID in the available building data for Exeter. If you have another OSID or a different query, please let me know!

host_agent

Output: No building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' was found within the Exeter area in the available building data.

If you have another OSID, a different location, or would like to clarify your request, please let me know how you'd like to proceed!